

Joint Optimization of Data Acquisition and Trajectory Planning for UAV-Assisted Wireless Powered Internet of Things

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Abstract—The development of Internet of Things (IoT) technology has led to the emergence of a large number of Intelligent Sensing Devices (ISDs). Since their limited physical sizes constrain the battery capacity, wireless powered IoT networks assisted by Unmanned Aerial Vehicles (UAVs) for energy transfer and data acquisition have attracted great interest. In this paper, we formulate an optimization problem to maximize system energy efficiency while satisfying the constraints of UAV mobility and safety, ISD quality of service and task completion time. The formulated problem is constructed as a Constrained Markov Decision Process (CMDP) model, and a Multi-agent Constrained Deep Reinforcement Learning (MCDRL) algorithm is proposed to learn the optimal UAV movement policy. In addition, an ISD-UAV connection assignment algorithm is designed to manage the connection in the UAV sensing range. Finally, performance evaluations and analysis based on real-world data demonstrate the superiority of our solution.

Index Terms—Constrained Markov decision process, data acquisition, UAV, wireless power transfer.

I. INTRODUCTION

DUE to the large-scale promotion and application of the Internet of Things (IoT) technology, a variety of rich IoT devices are springing up [1]. It is predicted that by 2030, Intelligent Sensing Devices (ISDs) will reach 500 billion units more [2]. However, such anytime and anywhere, a large number

of different wireless network connections also put tremendous pressure to the traditional terrestrial communication infrastructure, requiring not only high data transmission rates, but also large sensing ranges. In scenarios of data sensing, the effectiveness and timeliness of collected data become very important, such as health data monitoring for patients with chronic diseases in telemedicine, temperature and humidity monitoring of crop growth environment in agricultural production [3], and electrical fault monitoring of power lines and power towers in smart grids [4]. In these scenarios, the data generated in the corresponding ISDs needs to be collected for further processing and analysis, and thus effective data acquisition can increase the accuracy of derived decisions. However, the limited battery capacity of each ISD only supports its operation for a short period, and a low-powered ISD may lead to large transmission delay in data uploading, which has become a key constraint for realizing sustainable communications in the IoT.

The emergence of Wireless Power Transfer (WPT) technology provides a low-maintenance and flexible way to supply energy for ISDs [5]. By integrating WPT with Wireless Information Transfer (WIT) technology, it is possible to carry energy and information at the same time, i.e., transmitting energy to the ISD in the downlink and then collecting the monitoring data generated by the ISD in the uplink. In order to ensure energy-efficient WPT, it is necessary to build Line-of-Sight (LoS) links or improve communication channel conditions (e.g., reducing the distance among energy transfer devices) [6]. With the advantage of flexibility and mobility, Unmanned Aerial Vehicles (UAVs) are regarded as useful tools by carrying specific communication and energy transmitter units to improve transmission efficiency [7], compensating for the limited communication range of traditional IoT infrastructures [8].

The UAV-assisted energy transmission and data acquisition process is usually divided into three phases, i.e., the UAV moves to adjust its position, then connects to ISDs to provide energy within its communication range, and finally collects data from these devices. In general, the following challenges are faced:

- It is quite challenging to meet the data acquisition needs of widely distributed ISDs, which are limited by their physical sizes, limited battery capacities and computational processing capabilities. Thus, stable energy supply needs to be provided for energy-constrained ISDs to supplement their energy consumption for data uploading.

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- The sensing range of UAVs is limited by their three-dimensional (3D) locations, and it is difficult to regulate the location of UAVs to efficiently sense ISDs while reducing the path loss of communication channels.
- It is also challenging to minimize the distance among UAVs under the condition of safe flight, i.e., avoiding collision among multiple UAVs.

A community-based delay approximation algorithm was designed in [9] to quantify auction bids, and an auction algorithm is proposed to determine trajectories of UAVs. In [10], the authors investigated the beam fugitive vector trajectory planning of UAVs and the Central Processing Unit (CPU) frequency adaptation problem of ISDs, aiming to minimize system energy consumption. In [11], a model for UAV-assisted multi-carrier wireless powered communication is proposed, which is solved by jointly optimising the allocation of UAV trajectory, subcarrier, power and subslot to maximize the minimum achievable rate. In [12], an alternating optimization algorithm was developed aiming at jointly optimizing UAV trajectory planning, computational resource and bandwidth allocation. However, the above studies focusing on UAV position regulations only consider a 2D planar space and do not take into account the height of the UAV, i.e., a 3D space. In addition, only a single UAV is considered in the above schemes, and the position regulation of multiple UAVs is much more complicated. Moreover, the flight safety problem for UAV deployment is rarely considered in existing studies.

Therefore, there is an urgent need to design a stable energy transfer and reliable data acquisition scheme in energy-constrained IoT networks with various ISDs. To solve the above challenges, this paper focuses on multi-UAV assisted energy transfer and data acquisition, and considers their flight safety and 3D coordinate adjustment in a wireless-powered IoT network. The main contributions of this paper are summarized as follows:

- We construct a UAV-assisted wireless powered IoT system with the objective of maximizing system energy efficiency while ensuring UAV safety. A multi-objective optimization problem, with UAV mobility and ISD quality of service constraints, is formulated to make a trade-off between energy consumption and flight safety among UAVs. The formulated problem is decomposed into two sub-problems, i.e., UAV trajectory planning and ISD-UAV connection assignment.
- For the first subproblem, we propose a Multi-agent Constrained Deep Reinforcement Learning (MCDRL) algorithm. Specifically, to guarantee the safety of UAV flight while satisfying the optimization objective, the subproblem is reformulated as a Constrained Markov Decision Process (CMDP) problem, in which the safety constraint is viewed as a penalty constraint. In addition, it integrates actor-critic networks with personalized attention mechanisms, and adjusts different attention weights and penalty constraints to obtain the suitable UAV movement strategy.
- For the second subproblem, we propose a low-complexity ISD-UAV connection assignment algorithm based on matching theory. By maximizing the system energy efficiency of each time slot, the UAV-ISD connection is

determined and the quality of service as well as energy consumption constraints of ISDs are satisfied.

- The network distribution of ISDs is simulated based on a real-world dataset of Melbourne (Australia). We first evaluate the convergence of the designed algorithm, and then demonstrate its superiority in several metrics compared to representative schemes.

The rest of this paper is organized as follows. Section II discusses the related work. Section III illustrates the system model. Section IV formulates the optimization problem. Section V specifies our solution. Performance evaluations are presented in Section VI. Finally, we conclude this paper in Section VII.

II. RELATED WORK

In this section, we review the current studies related to UAV-enabled energy transfer and UAV trajectory optimization.

A. UAV-Enabled Energy Transfer

In recent years, UAV-assisted WPT has become a promising technology for wireless networks, by providing a sustainable energy supply for low-power sensors and IoT devices, to prolong their service time. Compared with traditional fixed-position energy transmitters, UAVs are more flexible.

Authors in [13] and [14] studied the time allocation problem in UAV-assisted WPT networks. Yan et al. [13] considered the UAV as an energy emitter and a data collector, formulated the problem of maximizing the amount of offloaded data in a given time, and derived the closed form expression of the optimal time allocation by analyzing the transmission efficiency. Du et al. [14] designed a time-division multiple access based workflow model, aiming to jointly optimize IoT device connection, computational resource allocation, UAV hovering time, WPT time and service sequence.

Authors in [15] and [16] studied the energy efficiency problem in UAV-assisted WPT networks. Wang et al. [15] considered the joint optimization of UAV location and transmit power allocation, formulated the energy efficiency maximization problem, and constructed the supermodular game model. Yan et al. [16] formulated the optimization problem of maximizing the total energy received by all sensors to obtain the best strategy for UAV deployment.

B. UAV Trajectory Optimization

In UAV-assisted wireless communication networks, UAVs make real-time decisions by sensing their surroundings, and thus the trajectory optimization of UAVs is particularly important. The studies of [17] and [18] used clustering techniques to determine the hovering position of UAVs, and then leveraged learning-based algorithms to determine their trajectories. Dai et al. [17] designed a hybrid heuristic and learning-based scheduling algorithm to solve the energy-efficient scheduling problem by jointly optimizing UAV trajectories, and computation as well as transmission resource allocations.

However, the approach of approximating the trajectory of a UAV through a limited number of path segments leads to the problem of excessive computational complexity [18]. For this reason, Guo et al. [19] designed a path discretization scheme to ensure high trajectory discretization accuracy while developing a path compression scheme to decompose the 3D path of the UAV into three 1D sub-paths. Pandiyan et al. [20] proposed a dynamic charging scheduling algorithm that combines genetic weighted clustering and nearest neighbor searching to schedule the optimal charging path for UAVs. In [21], the authors considered UAVs with different hover heights for hierarchical deployment and proposed a two-stage joint optimization algorithmic solution.

The studies in [2], [22], [23], [24] used deep reinforcement learning algorithms that allow the UAV to interact with the environment with local observations. In [2], Yu et al. designed an extended multi-objective deep reinforcement learning algorithm to learn online path planning strategies for UAVs based on demand priorities. Yang et al. [22] combine the problems of device selection, UAV placement and resource management to propose an asynchronous advantage actor-critic algorithm to improve the speed and accuracy of joint convergence. Wang et al. [23] proposed a trajectory-based block coordinate descent strategy and a deep deterministic policy-based gradient based control algorithm, and introduced the preferred experience replay technique to improve the convergence of the training process. Zhao et al. [24] formulated the trajectory planning problem as two coupled multi-agent stochastic games, and designed a decentralized reinforcement learning algorithm to decouple the problem and find its equilibrium point regarded as the optimal trajectory.

However, the above studies, involving reinforcement learning approaches, intend to maximize the overall reward by taking actions in the environment, while lacking the consideration of some non-reward factors and explicitly indicating constraints. In contrast, constrained deep reinforcement learning methods consider other important factors in addition to maximizing the reward, depending on the particular application domain or the particular task. In [25], Khairy et al. investigated joint dynamic multi-UAV height control and random channel access management, and proposed an online model-free constrained deep reinforcement learning algorithm to obtain an optimal wireless management control policy and improve the access efficiency of the network. Nevertheless, this study only considered the moving distance decision in one dimension of UAV height, and did not consider the 3D position adjustment decision of UAVs.

Nowadays, most of the articles focus on the position adjustment of single UAV. Yang et al. [26] investigated the minimisation of the total energy consumption of UAVs while achieving the minimum data transfer of the user by using dyadic and successive convex approximations, In order to obtain the optimal access order and trajectory. Yuan et al. [27] presented a UAV trajectory scheme constrained by the maximum flight speed, in order to minimize the energy consumption of the UAV and maximize the minimum harvested energy for the user, simultaneously. However, multi-UAV position adjustment is generally focused on 2 dimensions. Wang et al. [28] considered non-cooperative



Fig. 1. Illustration of system architecture.

multi-UAV scenarios, proposed a trajectory optimization problem to minimize the task completion time, and devised a decentralised deep reinforcement learning method to solve it. In practical applications, 3D position adjustment involving multiple UAVs is much complicated. In addition, traditional optimization methods of approximating UAV trajectories by a limited number of path segments lead to the problem of excessive computational complexity. Furthermore, they rarely consider flight safety issues for UAV applications, such as avoiding flight collisions during UAV operation.

Motivated by the observations, our study considers the data acquisition and trajectory planning problems in UAV-assisted wireless powered networks. However, previous related work has not taken into account the problem that some ISDs are rarely accessed or never sensed by UAVs, caused by the differences in the geographic distribution of devices, and the flight safety factors of multiple UAVs in the process of adjusting their movement positions.

III. SYSTEM MODEL

Fig. 1 shows an urban environment with multiple UAVs and ISDs, whose sets are denoted by $\mathcal{U} = \{1, 2, \dots, U\}$ and $\mathcal{I} = \{1, 2, \dots, I\}$, respectively. Due to the increasing demand for data acquisition and the limited battery capacity of ISDs, we consider a wireless power transmission and data acquisition system including multiple UAV-assisted ISDs. In the beginning, UAVs get sufficient energy from a charging station and move in a fixed boundary target area. The 3D coordinates of UAVs can be represented by $q_u(t) \in \mathbb{R}^{3 \times 1}$, where $q_u(t) = (x_u(t), y_u(t), h_u(t))$, and the position coordinates of ISDs are represented by $g_i(t) \in \mathbb{R}^{3 \times 1}$, where $g_i(t) = (x_i(t), y_i(t), 0)$. Due to the safety consideration of UAV flight, UAVs cannot collide with each other in the target area.

As shown in Fig. 2, the scheduling range is divided into T time slots, i.e., $\mathcal{T} = \{1, 2, \dots, T\}$. At the beginning of each time slot, the UAV is assumed to make position adjustment, and its corresponding position can be approximated as unchanged during this period. Then, the UAV transmits energy to ISDs

TABLE I
MAIN NOTATIONS

Notation	Definition
$i, I, \mathcal{I}$	Index, number, and set of ISDs.
$u, U, \mathcal{U}$	Index, number, and set of UAVs.
$t, T, \mathcal{T}$	Index, number, and set of time slots.
$q_u(t), g_i(t)$	Position coordinates of u -th UAV and i -th ISD in t -th time slot.
$F_i(t)$	i -th ISD task in t -th time slot.
$\chi_u(t), d_u(t)$	Flying action and movement distance of u -th UAV in t -th time slot.
d^{min}	Minimum safe flight distance.
$l^z, \hat{z}$	Sensing distance and range of UAV.
$l_{i,u}(t)$	Distance between u -th UAV and i -th ISD in t -th time slot.
$\alpha_{i,u}(t)$	User connection assignment between i -th ISD and u -th UAV in t -th time slot.
r_u	Channel gain of u -th UAV.
$e_u^m(t), e_u^w(t), e_u^c(t)$	Energy for movement, WPT, and data acquisition of u -th UAV in t -th time slot.
p^m, p^t, p^c	Movement power, transmit power and data acquisition power of UAV.
$\nu_u^m(t), \nu_u^w(t), \nu_u^c(t)$	Movement time, WPT time, data acquisition time of u -th UAV in t -th time slot.

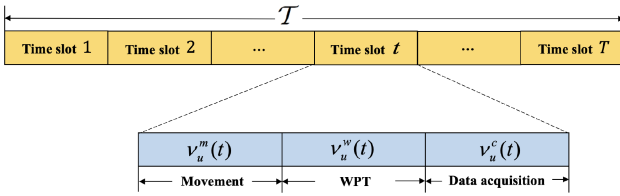


Fig. 2. Illustration of time architecture.

within its coverage area and collects the uploaded data from devices, spending $\nu_u^m(t)$, $\nu_u^w(t)$ and $\nu_u^c(t)$ time in turn.

Within a time slot, the data acquisition process can be divided into three steps: (1) the UAV moves to adjust its position and coverage area; (2) the UAV serves ISDs within its coverage area and transmits energy for devices to ensure they have sufficient energy for data uploading; and (3) the UAV collects the data uploaded by devices. The total time consumption of these three phases is defined as the duration of the time slot. Define each ISD has a latency-sensitive task represented by $F_i(t) = \{f_i(t), \nu^{max}\}$, where $f_i(t)$ denotes the size of the raw data (in bits) uploaded by ISD i . Since devices consume energy to upload data, the harvested energy of the device from the UAV is closely related to $f_i(t)$. Variable ν^{max} indicates the maximum cut-off time of the task, after which the uploaded data by devices is invalid. Main notations used in this paper are listed in Table I.

A. UAV Movement Model

Each UAV can move in certain direction $\chi_u(t) \in X \triangleq \{o^l, o^r, o^f, o^b, o^a, o^n, o^s\}$. Variables in order indicate horizontal left, horizontal right, horizontal forward, horizontal backward, vertical upward, vertical downward and stationary. The change in the UAV position of adjacent time-slots can be expressed by $d_u(t) = \|q_u(t+1) - q_u(t)\|$ [29]. Assuming that UAVs move

at constant speed η for simplicity, we can compute the duration time of the UAV movement by:

$$\nu_u^m(t) = \frac{\|q_u(t+1) - q_u(t)\|}{\eta}. \quad (1)$$

According to [30], the flight propulsion power of the UAV can be obtained by:

$$p^m(\eta) = p_o \left(1 + \frac{3\eta^2}{U_p^2}\right) + p_s \left(\sqrt{1 + \frac{\eta^4}{4V_h^4}} - \frac{\eta^2}{2V_h^2}\right)^{\frac{1}{2}} + \frac{1}{2}d_0\rho_a r_s J_r \eta^3, \quad (2)$$

where p_o and p_s are the blade profile power and the induced power, respectively. Variable U_p is the tip speed of the rotor, and V_h is the average induced speed of the rotor. Symbol d_0 is the airframe drag ratio, and ρ_a is the air density. Variable r_s is the rotor solidity, and J_r is the rotor disk area. Thus, the energy consumption of UAV movement is $e_u^m(t) = p_m \cdot \nu_u^m(t)$.

Each UAV can only move within the target area with the side lengths of x_u^{max} and y_u^{max} . In addition, h_u^{min} and h_u^{max} denote the minimum and maximum flight height of UAVs. The constraint ranges of UAV moving positions are:

$$0 \leq x_u(t) \leq x_u^{max}, \quad (3)$$

$$0 \leq y_u(t) \leq y_u^{max}, \quad (4)$$

$$h_u^{min} \leq h_u(t) \leq h_u^{max}. \quad (5)$$

Since the distance between any two UAVs should be more than the minimum safe flight distance, the following constraint should be satisfied:

$$\|q_u(t) - q_{u'}(t)\| \geq d_u^{min}. \quad (6)$$

where $q_{u'}(t)$ and d_u^{min} represent the position and minimum safe flight distance of UAVs other than UAV u , respectively.

B. ISD Charging Model

After the UAV moves to adjust its position, it is also assumed that its coverage area is located directly below it as in [31], and its horizontal coverage depends on the azimuth of the antenna and the altitude of the flight. The sensing distance and range in the horizontal coverage area of the UAV are defined as [31]:

$$l^z = h_u(t) \tan\left(\frac{\tau^{max}}{2}\right), \quad (7)$$

$$\hat{z} = \pi(l^z)^2 \sin\left(\frac{\tau^{max}}{2}\right)^2, \quad (8)$$

where l^z is its sensing distance, $\hat{z}$ is the sensing range of the UAV, and τ^{max} is the aperture angle. The three-dimensional distance between UAV u and ISD i can be calculated by:

$$l_{i,u}(t) = \sqrt{(q_u(t) - g_i(t))^2}. \quad (9)$$

Each sensing device can choose whether to be served by the UAV or not, i.e., $\alpha_{i,u}(t) = \{0, 1\}$, $\forall i \in \mathcal{I}, u \in \mathcal{U}, t \in \mathcal{T}$, where $\alpha_{i,u}(t) = 1$ indicates that the i -th ISD is served by the u -th UAV in time slot t , and $\alpha_{i,u}(t) = 0$ vice versa. In addition, since each ISD can be served by only one UAV in one time slot, the UAV first transmits the energy to the ISD to ensure it can operate properly, and then the UAV collects data from the ISD. As a result, the following condition should be satisfied:

$$\sum_{u=1}^U \alpha_{i,u}(t) = 1. \quad (10)$$

Since an ISD can only be served within the sensing range of the UAV, it satisfies:

$$\alpha_{i,u}(t) l_{i,u}(t) \leq l^z. \quad (11)$$

By considering the free-space path loss channel model, the channel gain of the UAV is:

$$r_u = \beta_0 l_{i,u}^{-2}(t) = \frac{\beta_0}{(q_u(t) - g_i(t))^2}, \quad (12)$$

where β_0 denotes the channel power gain at a reference distance of one meter. Furthermore, based on the generalized radio frequency power transmission model [25], received power $p_{i,u}^r(t)$ of the ISD within sensing range $\hat{z}$ of the UAV can be calculated by the Friis transmission equation:

$$p_{i,u}^r(t) = \alpha_{i,u} p_t \frac{r_u \mu^2}{(4\pi)^2 l_{i,u}^2(t)}, \quad (13)$$

where p_t is the transmit power of the UAV. Symbol $\mu = c/f_c$ is the wavelength, where c and f_c represent the speed of light and the carrier frequency of the radio frequency wave transmitted by the UAV, respectively, and θ is the path loss index of the air-to-ground channel. Consequently, the energy harvested by the ISD from the UAV is $e_i^h(t) = p_{i,u}^r(t) \nu_u^w(t)$, and the consumed energy for transmission from UAV to ISD is $e_u^w(t) = p_t \cdot \nu_u^w(t)$.

C. Data Acquisition Model

After transmitting energy to the ISD, the UAV starts to collect the uploaded data from ISDs. Note that this paper ignores

communication interference among multiple UAVs because orthogonal frequency resource allocation is utilized, which can reduce the possibility of mutual interference, especially when the coverage areas of multiple UAVs overlap [32]. This means that the communication among UAVs can remain relatively independent in the frequency domain, even when the spatial coverage overlaps. In addition, the interference among ISDs based on Orthogonal Frequency Division Multiple Access (OFDMA) technology can be ignored when ISDs upload data [23]. This is because OFDMA technology can divide the entire bandwidth into multiple subcarriers, and assign different subcarriers to devices transmitting data to the same UAV at the same time. The data rate of the upload link between the ISD and the UAV is:

$$\omega_{i,u}(t) = \alpha_{i,u} B \log_2 \left(1 + \frac{p_n r_u}{\sigma^2} \right), \quad (14)$$

where B is the bandwidth of the wireless channel, and p_n is the transmit power of the ISD, and σ^2 is the noise power.

Considering that each ISD has a different size of raw data, and sensors with sampling frequency δ_c are equipped on the UAV. The UAV collects data from the ISD by robin strategy [33], and the expected size of the collected data can be calculated by:

$$\Omega_{i,u}(t) = \min\{\delta_c \cdot \nu_u^c(t), \varepsilon_i(t)\}, \quad (15)$$

where $\varepsilon_i(t)$ is the size of the remaining raw data, i.e.,

$$\varepsilon_i(t) = f_i(0) - \sum_{t'=1}^{t-1} \delta_c \cdot \nu_u^c(t'), \quad (16)$$

where $f_i(0)$ is the size of the original data that the i -th ISD intends to upload. The consumed energy of the UAV to collect data is $e_u^c(t) = p_c \cdot \nu_u^c(t)$, and symbol p_c is the data acquisition power. Since the total amount of data collected by the UAV equals to the data size effectively uploaded by the device, the consumed energy of the data uploaded by the device is:

$$e_i^c(t) = p_n \frac{\Omega_{i,u}(t)}{\omega_{i,u}(t)}. \quad (17)$$

The consumed energy of the ISD to upload data cannot exceed the energy obtained during the UAV energy transfer, i.e.,

$$p_n \frac{\Omega_{i,u}(t)}{\omega_{i,u}(t)} \leq p_{i,u}^r(t) \nu_u^w(t). \quad (18)$$

To sum up, the total energy consumption to complete the UAV energy transfer and data acquisition is:

$$e_u(t) = e_u^m(t) + e_u^w(t) + e_u^c(t), \quad (19)$$

where $e_u^m(t)$, $e_u^w(t)$ and $e_u^c(t)$ represent the energy consumption of UAV movement, energy transmission, and data acquisition, respectively. The total energy consumption of the UAV should be less than initial battery capacity $E_u(0)$ of the UAV, i.e.,

$$\sum_{t=1}^T e_u(t) \leq E_u(0). \quad (20)$$

Furthermore, each data acquisition task should be completed by the service deadline, i.e.,

$$\nu_u^m(t) + \nu_u^w(t) + \nu_u^c(t) \leq \nu^{max}. \quad (21)$$

IV. PROBLEM FORMULATION

In this section, we formulate the energy efficiency problem based on the constraints of UAV-assisted ISD energy transmission and effective data acquisition. Our optimization objective is to maximize energy efficiency of the UAV, which can be calculated by the ratio of the bit number of the transmitted data to the total energy consumption of the UAV.

A. Optimization Objective

The following four aspects are considered for our optimization objective:

- *Effective data acquisition*: The effective data acquisition ratio of the UAV can be measured by the ratio of the collected data to the size of the original data, i.e.,

$$\lambda_{i,u}^{\varphi} = \frac{\sum_{i=1}^I \gamma_{i,u}^{\varphi}}{\sum_{i=1}^I f_i(0)}, \quad (22)$$

where $\gamma_{i,u}^{\varphi} = \sum_{t=1}^T \sum_{u=1}^U \Omega_{i,u}^{\varphi}(t)$ denotes the total amount of data collected from ISD i for UAV trajectory planning policy φ . Since one of our optimization objectives is to maximize the data acquisition, it is related to decision variables $q_u(t)$ and $\alpha_{i,u}(t)$, making it rather complicated to obtain the optimal policy.

- *Service fairness*: We define the service fairness of the UAV according to the Jain fairness index [34] as follows:

$$\Lambda_i(t) = \frac{(\sum_{i=1}^I \gamma_{i,u}^{\varphi})^2}{I \sum_{i=1}^I (\gamma_{i,u}^{\varphi})^2}. \quad (23)$$

We can notice that when the value of $\Lambda_i(t)$ approaches 1, the UAV will collect data from all ISDs in a balanced way to ensure the fairness of their services. As a result, maximizing $\Lambda_i(t)$ is also one of our optimization objectives.

- *Energy consumption*: Since the battery energy of UAVs is limited, they should reduce energy consumption $e_u(t)$ as much as possible to meet the service demand of ISDs, i.e.,

$$\kappa_u(t) = \frac{\sum_{t=1}^T e_u(t)}{\sum_{u=1}^U E_u(0)}, \quad (24)$$

where variable $\kappa_u(t)$ is defined by the ratio of the UAV energy consumption to initial energy $E_u(0)$. Then, another optimization objective is to minimize energy consumption $\kappa_u(t)$.

- *Flight safety*: Considering the safety of UAV flight, the following avoidance constraints need to be satisfied. 1) inequations (3)–(5) ensure that UAVs do not fly out of the target area; 2) inequation (6) avoids collision during UAV flying by considering the minimum flight safety distance among UAVs; 3) inequation (20) prevents failure caused by low power of UAVs; 4) inequation (21) ensures the quality of service of the ISD.

Based on the constraints analyzed above, we formulate the UAV-assisted data acquisition and trajectory planning problem in the WPT system as a multi-objective optimization problem

that maximize effective data acquisition, maximize service fairness, and minimize energy consumption as follows:

$$\mathcal{P}_1 : \max_{\varphi, \alpha_{i,u}} \lambda_{i,u}^{\varphi}, \quad (25)$$

$$\max_{\alpha_{i,u}} \Lambda_i(t), \quad (26)$$

$$\min_{\varphi, \alpha_{i,u}} \kappa_u(t). \quad (27)$$

$$\text{s.t.} \quad (3)–(6), (10), (11), (18), (20), (21).$$

Herein, constraints (3)–(5) restrict the 3D position range of the UAV. Constraint (6) guarantees the safe distance among flying UAVs. Constraint (10) shows that an ISD can be served by only one UAV. Constraint (11) ensures the quality of service of the ISD, i.e., the spatial distance between the UAV and the served ISD is within perceptual range l^z of the UAV to meet communication requirements. Constraint (18) makes sure that the energy provided by the UAV to the ISD can compensate for the energy consumption for data uploading. Constraint (20) limits the sustainable use of the UAV with energy consumption less than initial battery capacity $E_u(0)$. Constraint (21) restricts the total delay not exceed maximum service deadline ν^{max} .

According to objectives (25), (26) and (27), we transform the optimization problem $\mathcal{P}_1$ into an energy efficiency maximization problem. After removing the constants from problem $\mathcal{P}_1$, the ratio of effective data acquisition to UAV energy consumption is defined as an energy efficiency coefficient, the meaning of which refers to the energy consumption required per unit of data acquisition and takes into account the service fairness indicator. This whole is defined as energy efficiency to further determine the optimization objective of this paper, and the formulated problem becomes Problem $\mathcal{P}_2$:

$$\mathcal{P}_2 : \max_{\varphi, \alpha_{i,u}} \sum_{t=1}^T \sum_{u=1}^U \sum_{i=1}^I \frac{\Omega_{i,u}^{\varphi}(t) \Lambda_i(t)}{e_u(t)}, \quad (28)$$

$$\text{s.t.} \quad (3)–(6), (10), (11), (18), (20), (21).$$

Lemma 1: The formulated optimization problem $\mathcal{P}_2$ is NP-hard.

The simplified version of problem $\mathcal{P}_2$ belongs to the NP-hard ensemble coverage problem, thus problem $\mathcal{P}_2$ is also NP-hard. The detailed proof can be found in Appendix A, available online.

Due to the high flexibility of UAVs, solving Problem $\mathcal{P}_2$ to obtain UAV trajectory planning policy φ and ISD-UAV connection assignment decision $\alpha_{i,u}$ has the following difficulties:

- Constraints (18) and (20) contain non-linear data transmission rates and are made quadratic by the Euclidean distance calculations in constraints (6) and (21).
- Policy φ directly affects constraints (3) (6). It determines UAV position $q_u(t)$, which affects the flight range and flight safety distance of each UAV. In addition, UAV position $q_u(t)$ is interdependent with constraints (18) and (20). It determines the communication distance between the UAV and the ISD, which affects the data transmission rate and the quality of the communication link.

- UAV position $q_u(t)$ and $\alpha_{i,u}(t)$ are coupled in constraints (11) and (18). They determine whether the communication distance between the UAV and the ISD exceeds l^z . Connection assignment $\alpha_{i,u}(t)$ is binary and constraint (10) depends directly on $\alpha_{i,u}(t)$. In addition, UAV safety is affected by constraint (20). It restricts the energy consumption of the UAV in exchange for acquisition data.

Overall, the above challenges make Problem $\mathcal{P}_2$ rather difficult to be solved. For this reason, we illustrate the transformation of the problem in the next section.

B. Problem Transformation

To decouple decision variables and solve Problem $\mathcal{P}_2$ effectively, we relax constraints (11) and (18) by allowing all ISDs within the UAV sensing range to upload their data simultaneously, i.e., removing decision variable $\alpha_{i,u}$ from the received power and data upload rate in (13) and (14), respectively. Furthermore, the system energy efficiency maximization problem for the whole time can be decomposed into energy efficiency maximization subproblems in each time slot. This approach allows the system to make decisions based on the actual situation to meet the real-time requirements. Then, Problem $\mathcal{P}_2$ can be transformed as follows:

$$\mathcal{P}_3 : \max_{\varphi} \sum_{t=1}^T \sum_{u=1}^U \sum_{i=1}^I \frac{\Omega_{i,u}^{\varphi}(t) \Lambda_i(t)}{e_u(t)}, \quad (29)$$

s.t. (3)–(6), (18), (20).

$$\mathcal{P}_4 : \max_{\alpha_{i,u}} \sum_{u=1}^U \sum_{i=1}^I \frac{\Omega_{i,u}^{\varphi}(t) \Lambda_i(t)}{e_u(t)}, \quad (30)$$

s.t. (10), (11), (18), (20), (29).

where (29) is expressed as an upper bound on the time allocation for energy transfer. When the trajectory planning policy of the UAV is given, we can effectively evaluate the total mission completion time and make the following theorem to obtain it.

Theorem 1: When a UAV trajectory planning policy φ of UAV u is given, the upper bound of time allocation for energy transfer is:

$$(\nu_u^w)^* \leq \frac{\nu_u^{max} - \nu_u^m}{1 + p_{i,u}^r(t) \omega_{i,u}(t) / p_n \delta_c}. \quad (31)$$

The optimal time allocation for energy transfer can be obtained naturally when determining UAV trajectory as well as deriving the optimal data acquisition time.

The detailed proof can be found in Appendix B, available online.

Note that we propose an energy transfer and data collection scheme in the UAV-assisted WPT network by decoupling two decision variables following two steps to solve Problem $\mathcal{P}_2$. First, we determine the UAV trajectory planning strategy by solving Problem $\mathcal{P}_3$, based on multi-agent constrained reinforcement learning, and integrate attention mechanism to focus on reward and avoidability constraints to ensure UAV safety. Second, we solve problem $\mathcal{P}_4$ to maximize the system energy efficiency at each time slot.

Problem $\mathcal{P}_4$ is much easier to be solved than problem $\mathcal{P}_2$ in deciding ISD connection and resource allocation given the trajectory planning policy of the UAV, for which we develop an ISD-UAV connection assignment algorithm. The details of these two algorithms are described in the next section.

V. MULTI-UAV COOPERATIVE CDRL FRAMEWORK

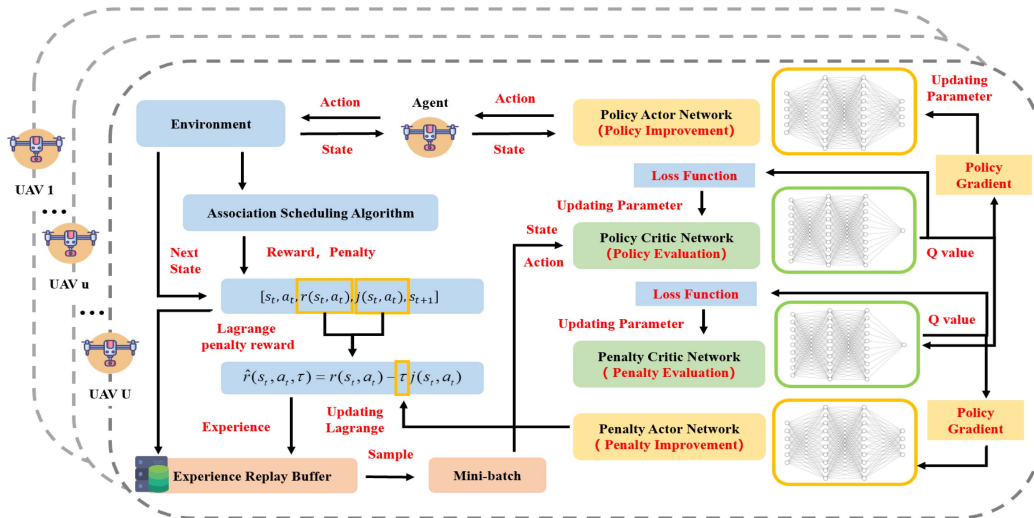
In this section, we solve the two subproblems in turn. To solve Problem $\mathcal{P}_3$ and find an optimal trajectory planning and connection assignment policy that satisfies the safety of UAV flight, we formulate it as a Constrained Markov Decision Process (CMDP) model and design an MCDRL algorithm by combining Deep Reinforcement Learning (DRL) with Lagrangian primal pairwise methods. Considering the optimization objective and satisfying the corresponding constraints require different levels of attention, so that agents learn to maximize the expected total discount reward and follow the constraints on the action simultaneously. Then, to solve Problem $\mathcal{P}_4$, an assignment algorithm is designed to determine the connection assignment of ISDs within the UAV sensing range to improve the efficiency of resource utilization.

A. CMDP Formulation

CMDP promises to be an alternative technique to enable Edge Learning (EL) that addresses the key challenges of Beyond 5G (B5G) networks. In [35], Xu et al. focused on providing a comprehensive overview of the state-of-the-art signal processing techniques for EL in B5G networks, not only pointing out the main EL techniques that can be used to develop B5G, but also providing the key open issues of B5G that have been raised during the application of the EL approach. CMDP is adaptive and flexible in the sense that it can flexibly adapt to changes in the environment and constraints to make adaptive decisions in a dynamic environment. This makes CMDP more advantageous when faced with complex and uncertain problems. At present, CMDP has been initially studied in the areas of semantic communication, edge computing and wireless sensing, and it is believed that CMDP can play an important role in B5G networks.

In this subsection, we construct a CMDP model for solving Problem $\mathcal{P}_3$, in which the agent's actions are limited by constraints and affect the interaction with the environment in turn. The CMDP model is composed of the following four key elements:

- **State:** Set $\mathcal{S}$ is the state space including $\mathcal{S} = \{s(t) = (S_1, S_2, S_3, S_4, S_5)\}$, $t \in \mathcal{T}$, where S_1 contains position coordinate $q_u(t)$ of UAV u , $\forall u \in \mathcal{U}$. Symbol S_2 contains the relative position of the UAV to the ISD, i.e., $\|q_u(t) - g_i(t)\|$, $\forall u \in \mathcal{U}, i \in \mathcal{I}$, and S_3 contains the relative position between UAV u and other UAVs, i.e., $\|q_u(t) - q_{u'}(t)\|$, $u \neq u' \in \mathcal{U}$. Symbol S_4 contains the remaining battery energy of UAV u , i.e., $E_u(0) - \sum_{t'=1}^t e_u(t')$, and S_5 includes the set of remaining transmission data of ISD i , i.e., $\varepsilon_i(t)$.
- **Action:** Set $\mathcal{A} = \{a(t) = a_u(t) = (\chi_u(t), d_u(t))\}$ is the set of joint actions. Variable $a_u(t)$ represents the movement of UAV u , which includes direction $\chi_u(t)$ of flight



movement and distance $d_u(t)$ of flight movement of UAV u in time slot t .

- *Reward*: After the UAV takes action $a(t)$ at state $s(t)$, the immediately received reward is defined by:

$$r_u(t) = \frac{\Omega_{i,u}(t)\Lambda_i(t)}{e_u(t)}. \quad (32)$$

- *Penalty*: The penalty constraint is denoted by $j(t)$ for UAV collision with other UAVs during the movement, i.e., the penalty for not satisfying (6).

The setup in the traditional CDRL uses a linear combination of original reward $r(t)$ and penalty constraint $j(t)$ to construct the final reward. However, weight values ϑ_m associated with penalty constraint are uncertain and need to be found by trial and error. In addition, the problem becomes rather difficult when there are multiple constraints. Therefore, we construct a Lagrangian penalty reward function as follows:

$$\hat{r}(s(t), a(t), \vartheta) = r(s(t), a(t)) - \sum_m \vartheta_m j(s(t), a(t)), \quad (33)$$

where ϑ is the Lagrange multiplier corresponding to the penalty constraint. In the following subsection, we propose an MCDRL algorithm to obtain reward $\hat{r}(s(t), a(t), \vartheta)$.

B. The MCDRL Algorithm

The proposed MCDRL algorithm uses a centralized training and distributed execution framework. In general, agents upload the observed states to the value network deployed at the central controller, and time difference errors are returned to agents during the training process. After completing the training, agents use the local policy networks to make decisions. Moreover, since obtaining the optimal policy and satisfying the constraints have different objectives, it is necessary to consider different attentions to the reward function and the penalty constraint. Therefore, each agent maintains $m + 1$ critic networks with attention mechanisms, i.e., a reward critic (associated with the

main reward function) and m penalty critic (associated with the penalty constraint). Herein, the attention mechanism allows the UAV agent to selectively focus on information about agents that are critical to its policy calculations. An evaluation factor is obtained by embedding information about the agents to measure the likelihood that remaining agents are important to each agent, rather than giving equal attention to all agents.

The MCDRL algorithm architecture and the pseudo-code of the algorithm are shown in Fig. 3 and Algorithm 1, respectively. Specifically, a CDRL agent is deployed on each UAV, which contains multiple deep neural networks, namely policy actor network ζ^π , policy critic network ζ^Q , and their respective target networks, i.e., $\zeta^{Q'} \leftarrow \zeta^Q$ and $\zeta^{\pi'} \leftarrow \zeta^\pi$. There are also penalty actor network ϕ^π , penalty critic network ϕ^Q , and their corresponding target network parameters, $\phi^{\pi'} \leftarrow \phi^\pi$ and $\phi^{Q'} \leftarrow \phi^Q$.

First, the algorithm starts by initializing the above network parameters, Lagrange multipliers $\vartheta_1, \dots, \vartheta_m$ and empirical replay buffer D . At the beginning of each epoch, all UAVs can serve the ISDs within their coverage area from different starting positions. By training the algorithm in multiple parallel environments, it is possible to improve the efficiency of sample use and reduce the variance of updates. Then, in time slot t , each UAV flies according to action $a(t)$, i.e., the direction and distance of UAV movement.

When the action execution of all UAVs is completed, the environment updates the state based on joint actions of UAVs to obtain state $s(t+1)$. Note that if UAV u flies out of the target area, its action is undone and it reverts to its previous state $s(t)$. Following this, we can obtain reward $r_u^e(t)$ of UAV u in parallel environment e and corresponding penalties $j_1^e, j_2^e, \dots, j_m^e$, and construct the Lagrangian penalty reward function by (30). Next, State transfer $(s_u^e(t), a_u^e(t), \hat{r}_u^e(t), j_1^e, j_2^e, \dots, j_m^e)$ is constructed and stored in the replay experience buffer D . When the buffer space reaches the limit, each UAV agent samples the experience in small batches in the replay experience buffer.

Algorithm 1: MCDRL Algorithm.

```

1: Initialize Actors: actor network parameter  $\zeta^\pi$ , penalty
   actor network parameter  $\phi^\pi$ 
2: Initialize Critics: critic network parameter  $\zeta^Q$ , penalty
   critic network parameter  $\phi^Q$ 
3: Initialize Targets: target network parameters  $\zeta^{\pi'} \leftarrow \zeta^\pi$ ,
    $\phi^{\pi'} \leftarrow \phi^\pi$ ,  $\zeta^{Q'} \leftarrow \zeta^Q$ ,  $\phi^{Q'} \leftarrow \phi^Q$ 
4: Initialize Lagrange multipliers  $\vartheta_1, \dots, \vartheta_m$ 
5: Initialize experience replay buffer  $D$ 
6: for  $epoch = 1, 2, \dots, L$  do
7:   Obtain initial observation  $s_u^e(t)$  of agent  $i$  in
   environment  $e$ 
8:   for  $t = 1, 2, \dots, T$  do
9:     Randomly sample action  $a_u^e(t)$ 
10:    Execute action and observe next state  $s_u^{*,e}$ 
11:    if UAV flies out of the boundary range then
12:      Terminate the UAV movement and set  $s_u^{*,e} = s_u^e$ 
13:    end if
14:    Obtain the ISD connection with UAVs using
    ISD-UAV connection assignment algorithm (i.e.,
    Algorithm 2)
15:    Obtain reward  $r_u^e(t), j_1^e, j_2^e, \dots, j_m^e$ 
16:    Let  $\hat{r}_u^e(t) = r_u^e(t) - \sum_{k=1}^m \vartheta_k j_k^e$ 
17:    Store
    ( $s_u^e(t), a_u^e(t), \hat{r}_u^e(t), j_1^e, j_2^e, \dots, j_m^e, s_u^e(t+1)$ ) into
    buffer  $D$ 
18:    if the experience replay buffer is full then
19:      Update parameters of the critic network  $\zeta^Q$  by
      minimizing its loss function according to (34)
20:      Update parameters of the penalty critic network
       $\phi^Q$  by minimizing its loss function according to
      (36)
21:      Update parameters of the actor network  $\zeta^\pi$  by
      using policy gradient approach according to (38)
22:      for  $k = 1, 2, \dots, m$  do
23:        Update Lagrangian multipliers
         $\vartheta_k \leftarrow \max(0, \vartheta_k + \iota(Q_{\phi^Q} - \varsigma_k))$ 
24:      end for
25:      Update target networks according to (40)
26:    end if
27:  end for
28: end for

```

When the experience replay buffer is full, we update the critic and actor networks, respectively. Then, we update lagrangian multipliers and target network parameters. Specifically, network parameter ζ^Q of the policy critic network is updated by minimizing loss function $\Delta(\zeta^Q)$, which is defined as follows:

$$\Delta(\zeta^Q) = \sum_{u=1}^U E[(Q_u(s(t), a(t)|\zeta^Q) - y_u(\zeta^Q))^2], \quad (34)$$

$$y_u(\zeta^Q) = \hat{r}_u^e(t) + \xi E[Q_u(s(t+1), a(t+1)|\zeta^Q) - \varsigma \log(\varphi_{\zeta^\pi}(a_u(t+1)|s_u(t+1)))]. \quad (35)$$

Algorithm 2: UAV and ISD Connection Assignment Algorithm.

```

1: Input:  $q_u(t), g_i(t), e_u^m(t), \langle f_i(t), \nu^{max} \rangle$ ;
2: Initialize connection assignment set
    $\mathcal{M}, \mathcal{C}_i = \emptyset, \forall i \in \mathcal{I}, \forall u \in \mathcal{U}$ ;
3: for UAV  $u = 1, 2, \dots, U$  do
4:   for ISD  $i = 1, 2, \dots, I$  do
5:     if (11) is met then
6:       Calculate energy transfer time  $\nu_u^w(t)$ ;
7:       Calculate  $e_u^w(t), e_u^c(t)$  and  $\Omega_{i,u}(t)$ ;
8:       Calculate  $\Omega_{i,u}/e_u(t)$  and store it into  $\mathcal{C}_i$ ;
9:     end if
10:   end for
11:   Sort the element in  $\mathcal{C}_i$  in a descending order;
12: end for
13: repeat
14:   for UAV  $u = 1, 2, \dots, U$  do
15:      $i = \text{GetTopItem}(\mathcal{C}_i)$ ;
16:     if (20) is met then
17:       if  $\mathcal{M}(i) = 0$  or  $(\Omega_{i,u}(t) \geq \Omega_{i,\mathcal{M}(i)}(t)$  and
        $(e_u(t) \leq e_{\mathcal{M}(i)}(t)))$  then
18:          $\mathcal{M}(i) = u$ 
19:       end if
20:        $\text{RemoveTopItem}(\mathcal{C}_i)$ 
21:     end if
22:   end for
23: until Each ISD in  $\mathcal{C}_i$  is associated.
24: output: Return  $\mathcal{M}$ 

```

Similarly, network parameters of the penalty critic are updated as follows:

$$\Delta(\phi^Q) = \sum_{u=1}^U E[(Q_u(s(t), a(t)|\phi^Q) - y_u(\phi^Q))^2], \quad (36)$$

$$y_u(\phi^Q) = \hat{r}_u^e(t) + \xi E[Q_u(s(t+1), a(t+1)|\phi^Q) - \varsigma \log(\varphi_{\phi^\pi}(a_u(t+1)|s_u(t+1)))], \quad (37)$$

where an entropy term is introduced in favor of the random strategy, and ς denotes the coefficient used to control strategy randomness.

Then, the policy actor network of agent u can be updated with parameter ζ_u^π using the policy gradient, which is defined as follows:

$$\begin{aligned} \Delta(\zeta_u^\pi) &= E[\nabla_{\zeta_u^\pi} \log(\varphi_{\zeta_u^\pi}(a_u(t)|s_u(t)))(-\varsigma \log(\varphi_{\zeta_u^\pi}(a_u(t)|s_u(t))) \\ &\quad + Q_{\zeta_u^Q}(s(t), a(t)) - b(s(t), a_{u'}(t)))] \end{aligned} \quad (38)$$

where b is a baseline function that is independent of action $a_u(t)$ of agent u , and $a_{u'}(t)$ represents actions of agents other than agent u .

Similar to the Lagrangian relaxation method, by updating the original and dual variables at different time scales, it is possible to guarantee that constraints are satisfied at the time of convergence. In view of this, after completing the parameter

updates of the policy critic network and the policy actor network on the fast time scale, the penalty actor network on the slower time scale can be updated with Lagrange multipliers in the following manner:

$$\vartheta_z \leftarrow \max(0, \vartheta_z + \iota(Q_{\phi Q} - \varsigma_k), \quad \forall z = 1, \dots, m, \quad (39)$$

where ι represents the learning rate for updating Lagrangian multipliers, and ς_k represents the threshold of the penalty constraint.

Finally, for respective target networks, the corresponding parameters can be updated by updating the rate with the following equation:

$$\zeta^{\pi'} \leftarrow \varrho \zeta^{\pi} + (1 - \varrho) \zeta^{\pi'}, \quad \zeta^{Q'} \leftarrow \varrho \zeta^Q + (1 - \varrho) \zeta^{Q'}, \quad (40)$$

where ϱ denotes the updating rate for network parameters of the target network. Similarly, the parameter update formulas for the penalty actor target network and the penalty critic target network are the same.

Lemma 2: The overall complexity of MCDRL algorithm in execution process is $\mathcal{O}((\sum_{p=1}^P n_p \cdot n_{p-1} + UI)T)$.

The time complexity of our solution depends mainly on Algorithms 1 and 2. The detailed proof can be found in Appendix C. So far, after obtaining the movement decision of the UAV, the UAV-ISD connection policy is determined by the UAV and ISD connection assignment algorithm proposed in the next subsection.

C. UAV and ISD Connection Assignment Algorithm

The pseudo code of the UAV and ISD connection assignment algorithm can be found in Algorithm 2. Its objective is to associate service with a UAV within its sensing range. Specifically, each ISD attempts to connect to a UAV that can improve system energy efficiency. In addition, each UAV selects the ISD according to the following standard conditions: 1) The ISD should be within its sensing range; 2) The ISD can obtain higher system energy efficiency, i.e., the larger the energy efficiency coefficient $\Omega_{i,u}/e_u(t)$, the higher priority given to the ISD when associating to the UAV.

First, we use set $\mathcal{M}$ with size I to record the connection assignment between ISDs and UAVs. If $\mathcal{M}(i) = u$, then the i th ISD is associated with the UAV. If $\mathcal{M}(i) = 0$, then the i th ISD has not been matched yet. In addition, we generate preference list $\mathcal{C}_u$ for UAV u to record ISDs that can increase energy efficiency. Constraint (11) is satisfied if the ISD is within the sensing range of the UAV, we can compute time $\nu_u^w(t)$ and consumption $e_u^w(t)$ of energy transfer, energy for data acquisition $e_u^c(t)$, and data acquisition size $\Omega_{i,u}(t)$. The ISDs with higher energy efficiency are stored in $\mathcal{C}_u$. Since UAVs need to collect more data and reduce energy as much as possible, the preferences are ranked to be relative to $\Omega_{i,u}(t)/e_u(t)$. It is possible that ISDs with higher energy efficiencies can be matched with higher priorities.

Second, each UAV keeps selecting ISDs based on its preference list and keeps checking whether the UAV's battery capacity can support its continued service according to constraint (20). Meanwhile, the selected ISD determines whether it matches the UAV. Condition $\mathcal{M}(i) = u'$ is updated if the selected ISD has



Fig. 4. A real-world map used in the simulation.

not been matched before, or if matching with UAV u' can achieve higher energy efficiency than the previous one. This procedure lasts until all ISDs in each preference list are checked. Then, the final ISD connection can be obtained from $\mathcal{M}$.

In general, we first propose an MCDRL algorithm with the aim of maximizing the energy efficiency of the system and taking into account the need for real-time decision making. In the MCDRL algorithm, the agent is trained by combining an actor-critic network with a personalized attention mechanism to obtain a trajectory planning policy for the UAV. Then, we propose a low-complexity UAV-ISD connection assignment algorithm to decide the UAV's connection management with ISDs within its sensing range.

VI. EXPERIMENTAL EVALUATIONS

In this section, we conduct simulation experiments based on real-world data to evaluate the effectiveness of our solution, and collect the geographic location of ISDs obtained from the IP addresses provided by APNIC [36] for the Melbourne CBD area in Australia. The online YouTube video service dataset [37] is used to simulate the data size of ISDs. First, we describe the simulation setup. Then, numerical results are given.

A. Simulation Setup

In our simulation, an Intel(R) Xeon(R) Silver 4210R CPU @ 2.40 GHz, python 3.7, and pytorch 1.1.0 are applied. As shown in Fig. 4, UAVs and ISDs (marked in red) are distributed in a rectangular area of 300 m $\times$ 700 m. In real scenarios, UAVs are usually restricted to operate at a certain altitude, e.g., in the U.S., UAVs can fly at 400 feet in the controlled airspace by obtaining LAANC or DroneZone authorization [34]. Therefore, we set the altitude range of UAVs between 50 m and 120 m [23]. Based on the physical dimensions of the UAV provided by DJI's website, the minimum safe distance among UAVs is set to be the length of wings, i.e., 5 m. The setting of simulation parameters can be found in Table II. For the network configurations during the training process, the number of hidden dimensions for both the critic network and the actor network is 128, and the number of training episodes is 1000. The learning rates of actor network, critic network, and Lagrange multiplier are 0.001, 0.001, and 0.0001 [25], respectively.

To verify the efficiency and effectiveness of the proposed algorithm, we consider the following three performance metrics:

TABLE II
SIMULATION PARAMETER SETTING

Parameters	Value
Bandwidth of wireless B	10MHz
Moving velocity μ_m	10m/s
Blade profile power p_o	79.86
Induced power p_s	88.63
Tip speed of the rotor U_p	120m/s
Average induced speed of the rotor V_h	4.03
Airframe drag ratio d_0	0.6
Air density ρ_a	$1.225kg/m^3$
Rotor solidity r_s	0.05
Rotor disk area J_r	$0.18m^2$
aperture angle ϑ_{max}	$\frac{\pi}{3}$
Reference channel power gain β_0	-50dB
Path loss index θ	2
Carrier frequency f_c	2.0GHz
Noise power σ^2	-110dBm
Sampling frequency δ_c	0.4

- *Data acquisition rate*: The total effective collected data divided by the generated data size of the ISD.
- *Service fairness*: According to (23), the average degree to which all UAVs collect data related to the target area when the current mission is completed.
- *Energy utilization*: The ratio of the total energy consumption divided by the initial battery energy of the UAV.

In order to evaluate the proposed algorithm, it is compared with the following three approaches:

- MADDPG [38]: It implements a policy for each agent based on a centralized policy critic network and a centralized penalty critic network. They are leveraged to estimate the value function of the penalty cost and multiple policy actor networks. Then, Lagrangian multipliers are improved by the penalty actor.
- MAAC [39]: It implements an unconstrained Multi-agent Actor-Attention-Critic (MAAC) algorithm, where no UAV agent is penalized for flying a safety distance.
- PPO [40]: It implements a Proximal Policy Optimization (PPO) algorithm, which is an improved algorithm for policy gradient reinforcement learning. For a fair comparison, we use the same deep neural network parameters to implement the PPO algorithm.

B. Performance Results

Figs. 5 and 6 demonstrate the comparison of the total training time for different update periods and different number of devices, respectively. In Fig. 5, since the UAV agent trajectories are updated at the beginning of each period to perform timely replacement of old data and avoid performance loss, it can be observed that the trend of the total training time variation tends to be smooth between two update periods, and MAAC

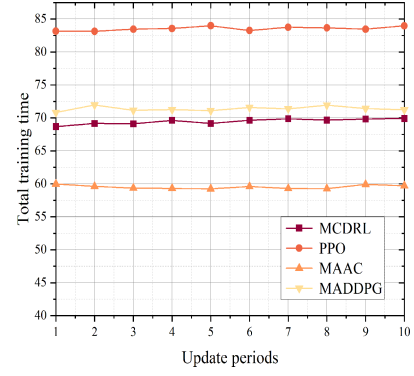


Fig. 5. Total training time for different update periods.

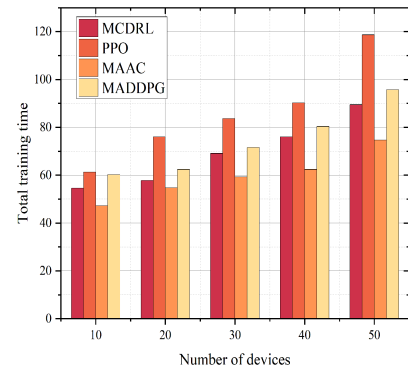


Fig. 6. Total training time for different number of devices.

has the shortest training time. In addition, the total training time of MCDRL is relatively shorter than that of MADDPG. In Fig. 6, the total training time increases with the number of devices, because the UAV agent takes longer time to explore the optimal trajectory planning and data collection strategies. For example, when the number of ISDs is 50, the total training time of MCDRL algorithm is 16.2% higher than that of MAAC algorithm, while it is 6.9% and 32.5% lower than that of MADDPG and PPO algorithms, respectively.

The changes in the loss, total reward, and penalty of different algorithms with the increasing number of training rounds are shown in Fig. 7, respectively. It can be observed that, as shown in Fig. 7(a), MCDRL, MADDPG, MAAC, and PPO converge gradually with the extension of the training process, while MCDRL has a better convergence speed. From Fig. 7(b), it can be seen that the total rewards of MCDRL and MADDPG algorithms increase, then decrease and level off with the increase of the training process. MAAC algorithm has the highest reward since there is no penalty constraint and the UAV does not consider the collision constraint for flight safety, and thus the total energy efficiency is higher. Nevertheless, MCDRL and MADDPG algorithms are constrained by the penalty constraint for safety flight distance, and their performance trends decrease until the penalty constraint is limited to the agreed range. As shown in Fig. 7(c), the penalty constraints in MCDRL and MADDPG algorithms increase and then decrease due to the moderating

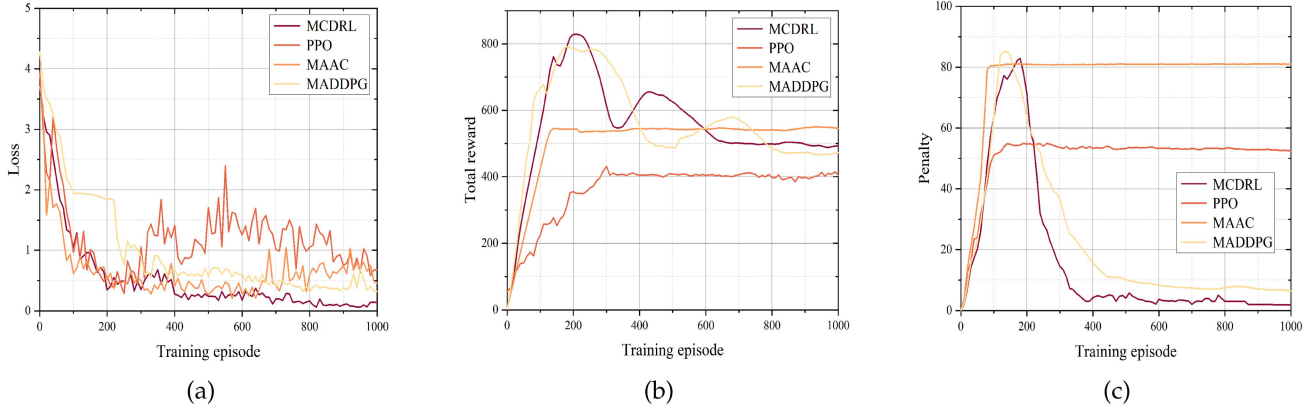


Fig. 7. Performance of algorithm during the training: (a) Loss, (b) Total reward, (c) Penalty.

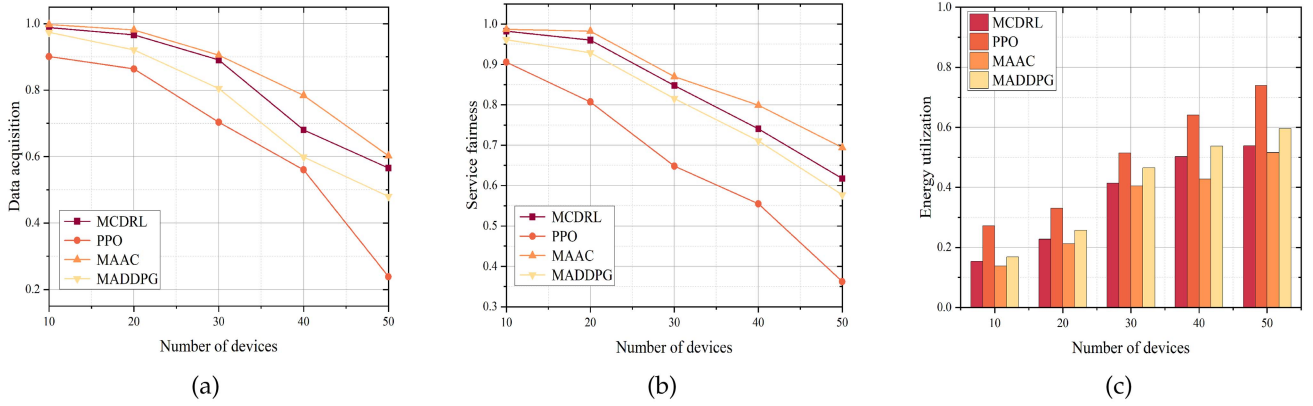


Fig. 8. Impact of the number of devices: (a) Data acquisition, (b) Service fairness, (c) Energy utilization.

effect of Lagrange multipliers in the constrained environment. However, MAAC and PPO algorithms have no constraint and their performance curves remain almost constant.

As shown in Fig. 8, the effects of different number of ISDs on effective data acquisition, service fairness, and energy utilization are compared. Specifically, when the number of ISDs increases, the total data volume increases due to the difference in data uploaded by ISDs, and therefore the collected data of UAVs decreases continuously. As shown in Fig. 8(a), when the number of ISDs is 30, data acquisition of MCDRL is slightly lower than that of MAAC algorithm. Similarly, in Fig. 8(b), the monotonic decrease in service fairness is with the increase of the number of devices. This is because when the number of UAVs is constant and the amount of data from ISDs increases, the service capacity of UAVs is limited to complete the efficient acquisition of data in the same duration. In addition, the UAV needs to consume more energy to meet the increasing service demand of ISDs, as shown in Fig. 8(c). The energy utilization of the UAV monotonically increases when the number of ISDs increases. The UAV needs to continuously fly and move to sense the location of ISDs. Thus, it consumes more transmission and data acquisition energy to accommodate the rich data of ISDs.

Fig. 9 illustrates the effects of different number of UAVs on data acquisition, service fairness, and energy utilization. As shown in Fig. 9(a) and (b), the effective data acquisition rate and service fairness increase with the increasing number of UAVs. This is because the sensing range and battery capacity of a single UAV are limited. With the same number of ISDs, deploying more UAVs can enhance the service capacity of the target area to meet energy transmission and data acquisition tasks of ISDs. It can be observed from Fig. 9(c) that the energy utilization tends to decrease when the number of UAVs increases, because a sufficient number of UAVs can provide a wide sensing range and adequate energy transmission resources as well as effective data acquisition services.

Fig. 10 illustrates the effect of different aperture angles on data acquisition, service fairness, and energy utilization. The aperture angles affect the sensing range of the UAV, which in turn impacts the service performance of the UAV. Specifically, in Fig. 10(a) and (b), data acquisition and service fairness show increasing trends with increasing aperture angles. At an aperture angle of 60 degrees, data acquisition of our solution MCDRL is slightly lower than that of the unconstrained MAAC algorithm, while higher than that of MADDPG and PPO by 11.7% and 25.9%, respectively. In terms of service fairness, MCDRL is higher than

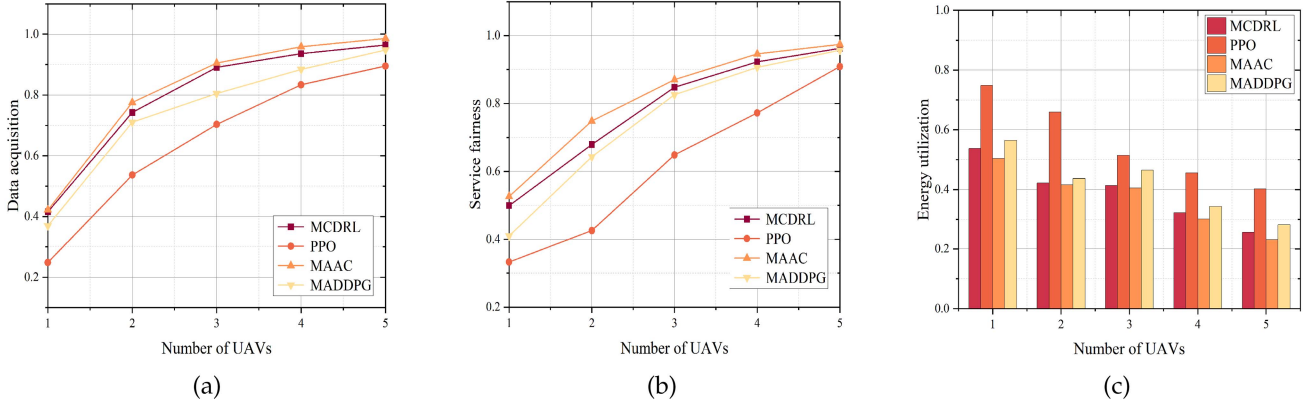


Fig. 9. Impact of the number of UAVs: (a) Data acquisition, (b) Service fairness, (c) Energy utilization.

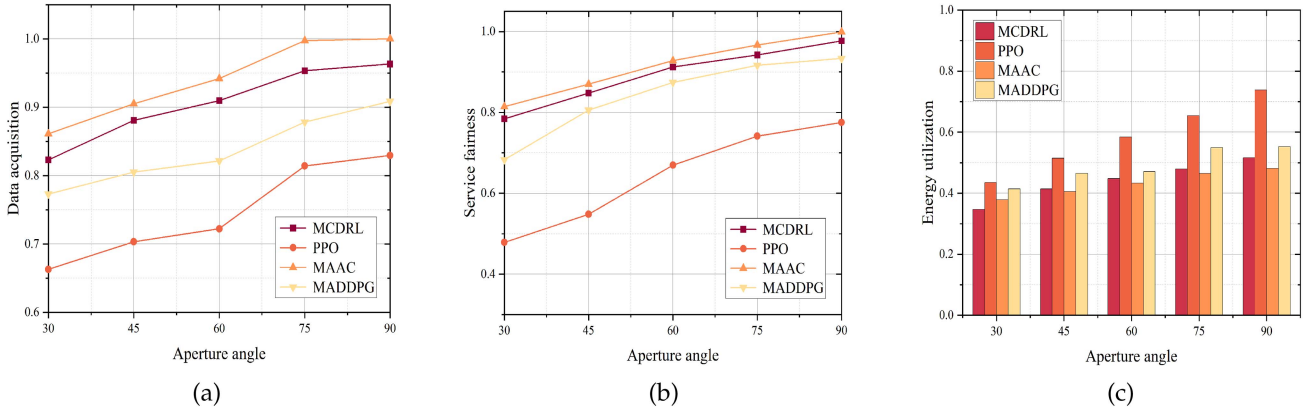


Fig. 10. Impact of aperture angle: (a) Data acquisition, (b) Service fairness, (c) Energy utilization.

that of MADDPG and PPO by 4.3% and 31.2%, respectively. In Fig. 10(c), the energy utilization tends to increase with the increasing aperture angles, because the UAV only needs to move with a shorter distance to sense more devices, and has more time to transmit energy and collect data.

VII. CONCLUSION

This paper establishes a wireless-powered IoT system for UAV-assisted energy transfer and data acquisition. Considering the UAV flight safety and energy consumption while ensuring the quality of service of ISDs, the problem of maximizing the system energy efficiency is formulated. It is further transformed into a UAV trajectory planning subproblem and an ISD-UAV connection assignment subproblem to decouple the UAV location adjustment and ISD connection assignment decision variables. Then, an MCDRL algorithm is developed and integrated with the attention mechanism to avoid collisions among UAVs, while UAVs provide services to ISDs to obtain the best trajectory planning strategy. In addition, an ISD-UAV connection assignment algorithm is proposed to determine the connection assignment policy with ISDs. Finally, simulation results show that, compared with representative solutions, our

designed algorithm has a shorter training time and is superior in terms of data acquisition, service fairness and energy utilization.

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